

CSA1713 – Artificial Intelligence

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Slot – C

10. A* ALGORITHM

Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("A* ALGORITHM")
graph = {}
n = int(input("Enter number of nodes: "))
for i in range(n):
    node = input("Node: ")
    graph[node] = {}
    e = int(input("Edges from " + node + ": "))
    for j in range(e):
        v = input("Connected node: ")
        c = int(input("Cost: "))
        graph[node][v] = c
h = {}
for i in graph:
    h[i] = int(input("Heuristic of " + i + ": "))
```

```
start = input("Start node: ")
```

```
goal = input("Goal node: ")
```

```
open = [start]
```

```
cost = {start: 0}
```

```
parent = {start: None}
```

```
while open:
```

```
    x = min(open, key=lambda i: cost[i] + h[i])
```

```
    open.remove(x)
```

```
    if x == goal:
```

```
        path = []
```

```
        while x:
```

```
            path.append(x)
```

```
            x = parent[x]
```

```
        print("Path:", path[::-1])
```

```
        break
```

```
    for y in graph[x]:
```

```
        g = cost[x] + graph[x][y]
```

```
        if y not in cost or g < cost[y]:
```

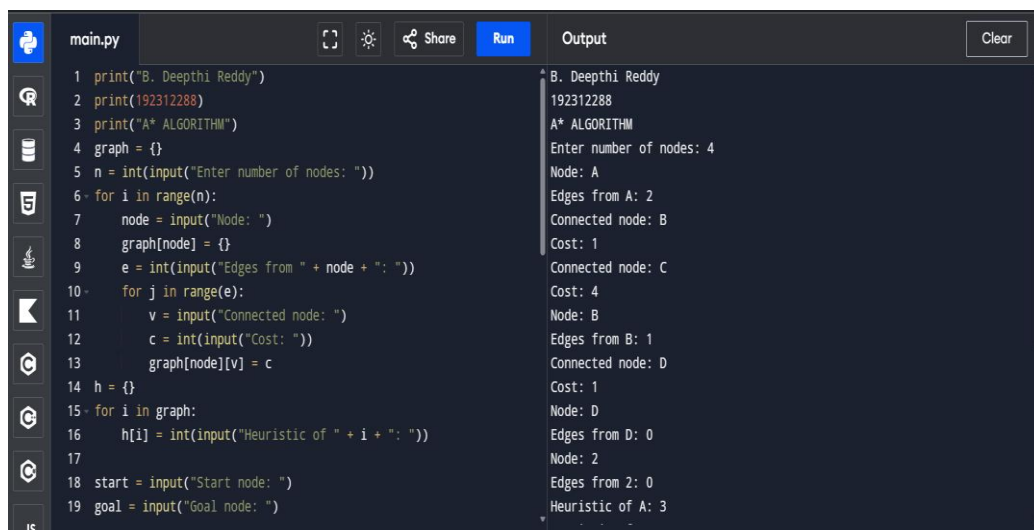
cost[y] = g

parent[y] = x

if y not in open:

open.append(y)

Output:



The image shows a code editor with a file named 'main.py' and an adjacent 'Output' window. The code implements an A* search algorithm. The output window displays the execution results, including the author's name, ID, algorithm name, and the step-by-step process of node expansion and heuristic calculation.

```
1 print("B. Deepthi Reddy")
2 print(192312288)
3 print("A* ALGORITHM")
4 graph = {}
5 n = int(input("Enter number of nodes: "))
6 for i in range(n):
7     node = input("Node: ")
8     graph[node] = {}
9     e = int(input("Edges from " + node + ": "))
10    for j in range(e):
11        v = input("Connected node: ")
12        c = int(input("Cost: "))
13        graph[node][v] = c
14 h = {}
15 for i in graph:
16     h[i] = int(input("Heuristic of " + i + ": "))
17
18 start = input("Start node: ")
19 goal = input("Goal node: ")
```

Output:

```
B. Deepthi Reddy
192312288
A* ALGORITHM
Enter number of nodes: 4
Node: A
Edges from A: 2
Connected node: B
Cost: 1
Connected node: C
Cost: 4
Node: B
Edges from B: 1
Connected node: D
Cost: 1
Node: D
Edges from D: 0
Node: 2
Edges from 2: 0
Heuristic of A: 3
```